

Program: Diploma in Fire Technology and Safety	
Course Code: 6461A	Course Title: Regulations for Safety, Health, and Environment
Semester: 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To understand the historical development and importance of Environmental Impact Assessment (EIA).
- To explore the tools, techniques, and methodologies for impact identification, prediction, and assessment in EIA.
- To learn about social impact assessment and documentation in EIA.
- To develop the skills needed for preparing EIA reports, Environmental Management Plans (EMP), and conducting post-project audits.
- To gain knowledge of key Indian regulations that govern safety, health, and environmental practices in industries.

Course Pre-requisites:

Topic	Course Code	CourseName	Semester
Principles of Safety Management		Principles of Safety Management	2
Electrical Safety		Electrical Safety	3
Environmental Safety		Environmental Safety	4
Safety in Engineering Industries		Safety in Engineering Industries	4

Course Outcomes

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Understand the historical development, legal, and regulatory aspects of EIA and key Indian regulations.	15	Understanding
CO2	Learn the techniques for impact identification and prediction in EIA.	14	Applying
CO3	Gain an understanding of social impact assessment and EIA documentation.	15	Understanding
CO4	Learn how to prepare an EIA report, develop Environmental Management Plans (EMP), and conduct audits.	14	Applying
	Series Test	2	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3	2					
CO3	3	3					
CO4	3	3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the historical development, legal, and regulatory aspects of EIA and key Indian regulations.		
M1.01	Historical development of Environmental Impact Assessment (EIA).	3	Understanding

M1.02	EIA in the project cycle and legal and regulatory aspects in India.	3	Understanding
M1.03	Understanding the types and limitations of EIA.	2	Understanding
M1.04	Process of EIA: screening, scoping, setting, analysis, and mitigation.	4	Applying
M1.05	Indian Regulations on Environmental Protection: Overview of key regulations.	3	Understanding
CO2	Learn the techniques for impact identification and prediction in EIA.		
M2.01	Various impact identification techniques: matrices, networks, and checklists.	3	Applying
M2.02	Learn about cost-benefit analysis and analysis of alternatives.	4	Applying
M2.03	Understand prediction tools and software packages for EIA.	4	Understanding
M2.04	Introduction to mathematical modeling for impact prediction.	3	Understanding
	Series Test I	1	
Contents: Methods of impact identification: matrices, networks, and checklists. Cost-benefit analysis and analysis of alternatives. Mathematical modeling and software tools for EIA prediction. Key Indian Regulations: National Green Tribunal (NGT) Act, 2010: Deals with environmental protection and the adjudication of environmental issues. The Hazardous Wastes (Management, Handling and Transboundary Movement) Rules, 2008: Governs the handling and disposal of hazardous waste in India.			
CO3	Gain an understanding of social impact assessment and EIA documentation.		
M3.01	Social impact assessment and its importance.	4	Understanding
M3.02	Relationship between social impacts and changes in community/institutional arrangements.	4	Understanding

M3.03	Individual and family level impacts and communities in transition.	3	Applying
M3.04	Documentation of EIA findings: planning, organization, and visual display.	4	Applying
Contents: Social impact assessment: definition and scope. Relationship between social impacts and community changes. Documentation techniques and preparation of the Environmental Impact Statement (EIS). Key Indian Regulations: The Factories Act, 1948: Ensures occupational health and safety. The Mines Act, 1952: Regulates the mining industry, ensuring health and safety standards.			
CO4	Learn how to prepare an EIA report, develop Environmental Management Plans (EMP), and conduct audits.		
M4.01	EIA report preparation and organization of information.	3	Applying
M4.02	Environmental Management Plan (EMP) preparation and implementation	4	Applying
M4.03	Mitigation and rehabilitation plans.	3	Applying
M4.04	Post-project audit and ethical aspects of EIA.	4	Understanding
	Series Test II	1	
Contents: Preparation of the EIA report. Development and implementation of the Environmental Management Plan (EMP). Mitigation and rehabilitation strategies. Post-project audit and ethical aspects of EIA. Key Indian Regulations: The National Environment Policy, 2006: Guidelines for sustainable development and environmental conservation in India. The Noise Pollution (Regulation and Control) Rules, 2000: Deals with noise pollution control in various industrial sectors.			

Text/Reference:

T/R	Book Title/Author
R1	Canter, L.W., Environmental Impact Assessment, McGraw Hill, New York,

	1996.
R2	Lawrence, D.P., Environmental Impact Assessment – Practical Solutions to Recurrent Problems, Wiley-Interscience, New Jersey, 2003.
R3	World Bank, Sourcebook on EIA.
R4	Marriott, Betty, Environmental Impact Assessment: A Practical Guide, McGraw Hill, 1997.
R5	The Environment Protection Act, 1986, Government of India.
R6	The Air (Prevention and Control of Pollution) Act, 1981, Government of India.
R7	The Water (Prevention and Control of Pollution) Act, 1974, Government of India.